

# IoT Cloud Platform

API Quick Reference · v2.0 · 2026-07-27

**Base URL:** `http://<server-host>:3000/api/v1` | **Auth:** `JWT = Authorization: Bearer <access_token>` | Key = `device_id + secret_key` in body/query | None = no auth required

**All responses:** `{ "success": true|false, "data": {...}, "error": {...} }` **Timestamps:** ISO-8601 (e.g. 2026-07-21T10:00:00.000Z)

**New in v2.0 — plan gating:** any request blocked by the pricing plan returns `error.code = "PLAN_LIMIT_EXCEEDED"` — HTTP 403 for a hard cap/disabled feature (nothing consumed), HTTP 429 for a monthly quota (resets next period). Every JWT-authenticated request also now counts against a plan-wide monthly API-call quota (not flagged per row below). Team members share the billing owner's plan, limits, and data.

Unchanged from v1.1 (37 endpoints)

Modified in v2.0 — plan check added  
(10 endpoints)

New in v2.0 (23 endpoints)

## 1. Health

| Method | Endpoint | Auth | Parameters | Returns                    |
|--------|----------|------|------------|----------------------------|
| GET    | /health  | None | —          | status, timestamp, version |

## 2. Authentication

| Method | Endpoint                      | Auth | Parameters                    | Returns                                                 |
|--------|-------------------------------|------|-------------------------------|---------------------------------------------------------|
| POST   | /auth/register                | None | email, password, display_name | user object, access_token, refresh_token, expires_in    |
| POST   | /auth/login                   | None | email, password               | user object, access_token, refresh_token, expires_in    |
| GET    | /auth/me                      | JWT  | —                             | user (id, email, display_name, is_verified, timestamps) |
| POST   | /auth/refresh                 | None | refresh_token                 | access_token, expires_in                                |
| POST   | /auth/logout                  | JWT  | refresh_token                 | success message                                         |
| GET    | /auth/verify-email            | None | ?token (query string)         | success message                                         |
| POST   | /auth/resend-verification     | None | email                         | generic success message (anti-enumeration)              |
| POST   | /auth/forgot-password         | None | email                         | generic success message (anti-enumeration)              |
| GET    | /auth/reset-password/validate | None | ?token (query string)         | email, expires_at                                       |
| POST   | /auth/reset-password          | None | token, new_password           | success message                                         |

## 3. Plans & Billing

Every user is auto-assigned the Free plan at registration via a DB trigger — no client action required.

| Method | Endpoint                 | Auth | Parameters                                                 | Returns                                                            |
|--------|--------------------------|------|------------------------------------------------------------|--------------------------------------------------------------------|
| GET    | /plans <b>NEW</b>        | None | —                                                          | array of all 5 plans with every limit field                        |
| GET    | /account/plan <b>NEW</b> | JWT  | —                                                          | effective plan, subscription status, current usage vs. every quota |
| PUT    | /account/plan <b>NEW</b> | JWT  | plan_code<br>(free basic standard professional enterprise) | updated plan_code,<br>current_period_start/end                     |

## 4. Team / Organization

Owner-only unless noted. A user can be an active member of at most one organisation at a time.

| Method | Endpoint                                   | Auth | Parameters                         | Returns                                                       |
|--------|--------------------------------------------|------|------------------------------------|---------------------------------------------------------------|
| POST   | /team/invitations <b>NEW</b>               | JWT  | email, role? (admin member viewer) | invitation object (id, email, role, expires_at)               |
| GET    | /team/invitations <b>NEW</b>               | JWT  | —                                  | array of pending invitations (owner-scoped)                   |
| DELETE | /team/invitations/:id <b>NEW</b>           | JWT  | :id (URL)                          | success message                                               |
| POST   | /team/invitations/:token/accept <b>NEW</b> | JWT  | :token (URL)                       | success message                                               |
| GET    | /team/members <b>NEW</b>                   | JWT  | —                                  | owner + array of active members, meta.limit                   |
| PUT    | /team/members/:memberId <b>NEW</b>         | JWT  | role (admin member viewer)         | updated membership                                            |
| DELETE | /team/members/:memberId <b>NEW</b>         | JWT  | :memberId (URL)                    | success message (owner removes anyone; member can self-leave) |

## 5. Projects

| Method | Endpoint             | Auth | Parameters          | Returns                               |
|--------|----------------------|------|---------------------|---------------------------------------|
| GET    | /projects            | JWT  | —                   | array of projects (with device_count) |
| POST   | /projects            | JWT  | name, description?  | created project object                |
| GET    | /projects/:projectId | JWT  | :projectId (URL)    | single project (with device_count)    |
| PUT    | /projects/:projectId | JWT  | name?, description? | updated project object                |
| DELETE | /projects/:projectId | JWT  | :projectId (URL)    | success message                       |

## 6. Devices

Team members now share access to the owner's projects/devices (previously single-owner only) — applies platform-wide, not flagged per row.

| Method | Endpoint                     | Auth | Parameters       | Returns                                                        |
|--------|------------------------------|------|------------------|----------------------------------------------------------------|
| GET    | /projects/:projectId/devices | JWT  | :projectId (URL) | array of devices (with status, sensor_count, firmware_version) |

| Method | Endpoint                                | Auth | Parameters          | Returns                                                                                                                                                                                                |
|--------|-----------------------------------------|------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST   | /projects/:projectId/devices <b>MOD</b> | JWT  | name, description?  | created device + secret_key (shown ONCE only)<br><i>now enforces plan device limit — 403 PLAN_LIMIT_EXCEEDED on Free plan past 2 devices; paid plans allow overage billed at extra_device_cost_inr</i> |
| GET    | /devices/:deviceId                      | JWT  | :deviceId (URL)     | single device (with status, sensor_count, firmware_version)                                                                                                                                            |
| PUT    | /devices/:deviceId                      | JWT  | name?, description? | updated device object                                                                                                                                                                                  |
| DELETE | /devices/:deviceId                      | JWT  | :deviceId (URL)     | success message                                                                                                                                                                                        |
| GET    | /devices/:deviceId/status               | JWT  | :deviceId (URL)     | status, ip_address, last_seen_at                                                                                                                                                                       |

## 7. Sensors

| Method | Endpoint                              | Auth | Parameters                 | Returns                                                                                              |
|--------|---------------------------------------|------|----------------------------|------------------------------------------------------------------------------------------------------|
| GET    | /devices/:deviceId/sensors            | JWT  | :deviceId (URL)            | array of sensors (with variable_count)                                                               |
| POST   | /devices/:deviceId/sensors <b>MOD</b> | JWT  | name, description?, unit?  | created sensor object<br><i>now enforces plan sensors-per-device limit — 403 PLAN_LIMIT_EXCEEDED</i> |
| GET    | /sensors/:sensorId                    | JWT  | :sensorId (URL)            | sensor with embedded variables[]                                                                     |
| PUT    | /sensors/:sensorId                    | JWT  | name?, description?, unit? | updated sensor object                                                                                |
| DELETE | /sensors/:sensorId                    | JWT  | :sensorId (URL)            | success message                                                                                      |

## 8. Variables

| Method | Endpoint                                | Auth | Parameters        | Returns                                                                                                  |
|--------|-----------------------------------------|------|-------------------|----------------------------------------------------------------------------------------------------------|
| GET    | /sensors/:sensorId/variables            | JWT  | :sensorId (URL)   | array of variables (latest_value, latest_string_value, latest_at)                                        |
| POST   | /sensors/:sensorId/variables <b>MOD</b> | JWT  | label, data_type  | created variable object<br><i>now enforces plan variables-per-sensor limit — 403 PLAN_LIMIT_EXCEEDED</i> |
| GET    | /variables/:variableId                  | JWT  | :variableId (URL) | variable with latest_value, latest_string_value, latest_at                                               |
| PUT    | /variables/:variableId                  | JWT  | label             | updated variable object                                                                                  |
| DELETE | /variables/:variableId                  | JWT  | :variableId (URL) | success message                                                                                          |

## 9. Command Templates & Dispatch

| Method | Endpoint                             | Auth | Parameters      | Returns                                     |
|--------|--------------------------------------|------|-----------------|---------------------------------------------|
| GET    | /devices/:deviceId/command-templates | JWT  | :deviceId (URL) | array of templates (each with parameters[]) |

| Method | Endpoint                                           | Auth | Parameters                                            | Returns                                                                                                                        |
|--------|----------------------------------------------------|------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| POST   | /devices/:deviceId/command-templates <b>MOD</b>    | JWT  | name, description?, parameters[]                      | created template with parameters[]<br><i>now enforces plan templates-per-device limit — 403 PLAN_LIMIT_EXCEEDED</i>            |
| GET    | /command-templates/:templateId                     | JWT  | :templateId (URL)                                     | template with parameters[]                                                                                                     |
| PUT    | /command-templates/:templateId                     | JWT  | name?, description?, parameters[]?                    | updated template (parameters[] = full replacement)                                                                             |
| DELETE | /command-templates/:templateId                     | JWT  | :templateId (URL)                                     | success message (also deletes all instances)                                                                                   |
| POST   | /command-templates/:templateId/dispatch <b>MOD</b> | JWT  | parameters{} (key-value), ttl_seconds? (default 3600) | command instance status=queued — HTTP 202<br><i>now enforces plan monthly command-dispatch quota — 429 PLAN_LIMIT_EXCEEDED</i> |
| GET    | /command-instances/:instanceId                     | JWT  | :instanceId (URL)                                     | instance with status, parameters{}, timestamps                                                                                 |
| GET    | /devices/:deviceId/command-history                 | JWT  | :deviceId, ?page, ?limit, ?status                     | paginated list (meta: total, page, total_pages)                                                                                |

## 10. Telemetry Queries

from and to are required for /telemetry and /telemetry/aggregate. Query window is now capped by the caller's plan (query\_range\_days) — 400 QUERY\_RANGE\_EXCEEDED.

| Method | Endpoint                                              | Auth | Parameters                                                              | Returns                                                                                                                                         |
|--------|-------------------------------------------------------|------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /variables/:variableId/telemetry <b>MOD</b>           | JWT  | :variableId, ?from (ISO), ?to (ISO), ?limit (1–5000, default 1000)      | readings[] with value and recorded_at. Max 5000 rows.<br><i>now enforces plan query-range limit — 400 QUERY_RANGE_EXCEEDED</i>                  |
| GET    | /devices/:deviceId/telemetry/latest                   | JWT  | :deviceId (URL)                                                         | latest value for every variable on device (incl. variable_id, sensor_id)                                                                        |
| GET    | /variables/:variableId/telemetry/aggregate <b>MOD</b> | JWT  | :variableId, ?from (ISO), ?to (ISO), ?interval (1m 5m 15m 30m 1h 6h 1d) | buckets[] with time, min, max, avg, count<br><i>now enforces plan query-range limit — 400 QUERY_RANGE_EXCEEDED</i>                              |
| GET    | /variables/:variableId/telemetry/export <b>NEW</b>    | JWT  | :variableId, ?from (ISO), ?to (ISO), ?format (csv json, default csv)    | file download (Content-Disposition: attachment), max 100,000 rows<br><i>gated by plan.export_csv / plan.export_json — Free plan has neither</i> |

## 11. Rules — Automation & Alerting Engine

Requires plan.rule\_engine\_enabled (Standard+) — 403 PLAN\_LIMIT\_EXCEEDED otherwise. Evaluated synchronously the instant a matching telemetry reading is ingested.

| Method | Endpoint                            | Auth | Parameters      | Returns                        |
|--------|-------------------------------------|------|-----------------|--------------------------------|
| GET    | /devices/:deviceId/rules <b>NEW</b> | JWT  | :deviceId (URL) | array of rules for this device |

| Method | Endpoint                            | Auth | Parameters                                                                                                                                                                                         | Returns                                                              |
|--------|-------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| POST   | /devices/:deviceId/rules <b>NEW</b> | JWT  | name, variable_id, operator (>,<,>=,<=,=,!=), threshold, action_type (dispatch_command send_alert), template_id?, action_parameters?, alert_channels?, alert_message?, cooldown_seconds?, enabled? | created rule object                                                  |
| GET    | /rules/:ruleId <b>NEW</b>           | JWT  | :ruleId (URL)                                                                                                                                                                                      | single rule object                                                   |
| PUT    | /rules/:ruleId <b>NEW</b>           | JWT  | any rule field (partial update)                                                                                                                                                                    | updated rule object                                                  |
| DELETE | /rules/:ruleId <b>NEW</b>           | JWT  | :ruleId (URL)                                                                                                                                                                                      | success message                                                      |
| GET    | /rules/:ruleId/logs <b>NEW</b>      | JWT  | :ruleId (URL)                                                                                                                                                                                      | trigger history: triggered_at, variable_value, action_result, detail |

## 12. Webhooks

Requires plan.webhooks\_enabled (Professional+) — 403 PLAN\_LIMIT\_EXCEEDED otherwise. URL must be https://.

| Method | Endpoint                            | Auth | Parameters                                                                                            | Returns                                                            |
|--------|-------------------------------------|------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| GET    | /webhooks <b>NEW</b>                | JWT  | —                                                                                                     | array of webhooks (no secret field)                                |
| POST   | /webhooks <b>NEW</b>                | JWT  | url (https://), event_types[] (device.online device.offline command.success command.failed), enabled? | created webhook incl. secret (shown ONCE only)                     |
| PUT    | /webhooks/:id <b>NEW</b>            | JWT  | url?, event_types[]?, enabled?                                                                        | updated webhook (no secret field)                                  |
| DELETE | /webhooks/:id <b>NEW</b>            | JWT  | :id (URL)                                                                                             | success message                                                    |
| GET    | /webhooks/:id/deliveries <b>NEW</b> | JWT  | :id (URL)                                                                                             | delivery log: event_type, payload, response_status, success, error |

## 13. Alerts History

| Method | Endpoint                   | Auth | Parameters    | Returns                                                               |
|--------|----------------------------|------|---------------|-----------------------------------------------------------------------|
| GET    | /alerts/history <b>NEW</b> | JWT  | ?page, ?limit | paginated log of email/WhatsApp sends (channel, trigger_type, status) |

## 14. Device Ingest (called by IoT devices — NOT the app)

readings[] item: variable\_id (UUID), value (any), recorded\_at? (ISO-8601, defaults to server time). Max 500 readings per call. Polling /commands/pending marks returned commands as 'sent' AND acts as a heartbeat update.

| Method | Endpoint                                | Auth | Parameters                                                      | Returns                                                                                                                                                                                                      | Rate Limit |
|--------|-----------------------------------------|------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| POST   | /ingest/telemetry<br>MOD                | Key  | device_id, secret_key, readings[]                               | inserted count, rejected count (HTTP 207 if partial failure)<br><i>now enforces plan monthly telemetry-message quota (429), and synchronously evaluates any matching rules + fires device.online webhook</i> | 600/min    |
| GET    | /ingest/commands/pending                | Key  | ?device_id, ?secret_key                                         | array of queued commands (auto-marked as sent)                                                                                                                                                               | 300/min    |
| POST   | /ingest/commands/:instanceId/ack<br>MOD | Key  | device_id, secret_key, status (success failed), failure_reason? | updated instance (id, status, executed_at)<br><i>now fires a command.success / command.failed webhook, and a command_failed alert on failure — request/response unchanged</i>                                | 120/min    |
| POST   | /ingest/heartbeat<br>MOD                | Key  | device_id, secret_key, firmware_version?, ip_address?           | status: online, device_id<br><i>now fires a device.online webhook on the offline→online transition — request/response unchanged</i>                                                                          | 120/min    |

## 15. Standard Error Codes

| Error Code                          | HTTP | When It Occurs                                               |
|-------------------------------------|------|--------------------------------------------------------------|
| VALIDATION_ERROR                    | 400  | Request body/params failed schema validation                 |
| WEAK_PASSWORD                       | 400  | Password does not meet strength requirements                 |
| SAME_PASSWORD                       | 400  | New password is identical to current password                |
| MISSING_PARAMS                      | 400  | Required query params (from, to) are missing                 |
| INVALID_INTERVAL                    | 400  | Aggregate interval not one of the allowed values             |
| MISSING_REQUIRED_PARAM              | 400  | A required command parameter was not supplied                |
| INVALID_PARAM_TYPE                  | 400  | Command parameter value does not match declared type         |
| UNSUPPORTED_TYPE                    | 400  | Aggregation requested on a string variable                   |
| INVALID_VERIFY_TOKEN                | 400  | Email verification link is invalid or expired                |
| INVALID_RESET_TOKEN                 | 400  | Password reset link is invalid, expired, or already used     |
| QUERY_RANGE_EXCEEDED                | 400  | Requested time window is wider than the caller's plan allows |
| INVALID_VARIABLE                    | 400  | Rule's variable_id does not belong to the target device      |
| INVALID_TEMPLATE / MISSING_TEMPLATE | 400  | Rule's template_id missing or does not belong to the device  |
| MISSING_CHANNELS                    | 400  | send_alert rule created with no alert_channels               |
| INVALID_URL                         | 400  | Webhook URL does not start with https://                     |
| INVALID_FORMAT                      | 400  | Telemetry export format is not csv or json                   |
| CANNOT_JOIN_OWN_ORG                 | 400  | Owner tried to accept their own team invitation              |

| Error Code                                                                                                                   | HTTP      | When It Occurs                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INVALID_INVITATION                                                                                                           | 400       | Team invitation token invalid, expired, or already used                                                                                                                       |
| UNAUTHORIZED                                                                                                                 | 401       | Missing or expired JWT access token                                                                                                                                           |
| TOKEN_REVOKED                                                                                                                | 401       | Access token was explicitly revoked (user logged out elsewhere)                                                                                                               |
| INVALID_CREDENTIALS                                                                                                          | 401       | Wrong email or password on login                                                                                                                                              |
| INVALID_REFRESH_TOKEN                                                                                                        | 401       | Refresh token invalid, expired, or already used                                                                                                                               |
| DEVICE_UNAUTHORIZED                                                                                                          | 401       | Device secret_key does not match stored hash                                                                                                                                  |
| NO_ACTIVE_SUBSCRIPTION                                                                                                       | 401       | No plan subscription found for the account (should never occur — auto-assigned at registration)                                                                               |
| NOT_BILLING_OWNER                                                                                                            | 403       | A team member tried to perform an owner-only action (billing, invites, member management)                                                                                     |
| PLAN_NOT_SELF_SERVE                                                                                                          | 403       | Tried to self-service switch to the Enterprise plan (sales-assisted only)                                                                                                     |
| EMAIL_MISMATCH                                                                                                               | 403       | Accepted a team invitation while logged in as a different account                                                                                                             |
| FORBIDDEN                                                                                                                    | 403       | Generic — e.g. a non-owner member tried to remove a different member                                                                                                          |
| PLAN_LIMIT_EXCEEDED                                                                                                          | 403 / 429 | The core plan-gating error — a hard resource cap or disabled feature (403, nothing consumed) or a monthly quota (429, resets next period). Always check error.details.metric. |
| PROJECT_NOT_FOUND / DEVICE_NOT_FOUND /<br>SENSOR_NOT_FOUND / VARIABLE_NOT_FOUND /<br>TEMPLATE_NOT_FOUND / INSTANCE_NOT_FOUND | 404       | Resource does not exist or belongs to another organisation (never distinguished, for security)                                                                                |
| RULE_NOT_FOUND / WEBHOOK_NOT_FOUND /<br>MEMBER_NOT_FOUND / INVITATION_NOT_FOUND /<br>PLAN_NOT_FOUND                          | 404       | Resource does not exist or belongs to another organisation                                                                                                                    |
| EMAIL_ALREADY_EXISTS                                                                                                         | 409       | Email is already registered                                                                                                                                                   |
| DEVICE_NAME_EXISTS                                                                                                           | 409       | Device name already used within this project                                                                                                                                  |
| LABEL_ALREADY_EXISTS                                                                                                         | 409       | Variable label already used on this sensor                                                                                                                                    |
| TEMPLATE_NAME_EXISTS                                                                                                         | 409       | Command template name already used on this device                                                                                                                             |
| DUPLICATE_PARAM_NAME                                                                                                         | 409       | Duplicate parameter name within a command template                                                                                                                            |
| ALREADY_IN_ANOTHER_ORG                                                                                                       | 409       | User must leave their current organisation before accepting a new invite                                                                                                      |
| ACCOUNT_LOCKED                                                                                                               | 429       | Account locked after 5 failed login attempts — check retry_after                                                                                                              |
| RATE_LIMIT_EXCEEDED                                                                                                          | 429       | Too many requests to this endpoint                                                                                                                                            |
| INTERNAL_ERROR                                                                                                               | 500       | Unexpected server error                                                                                                                                                       |